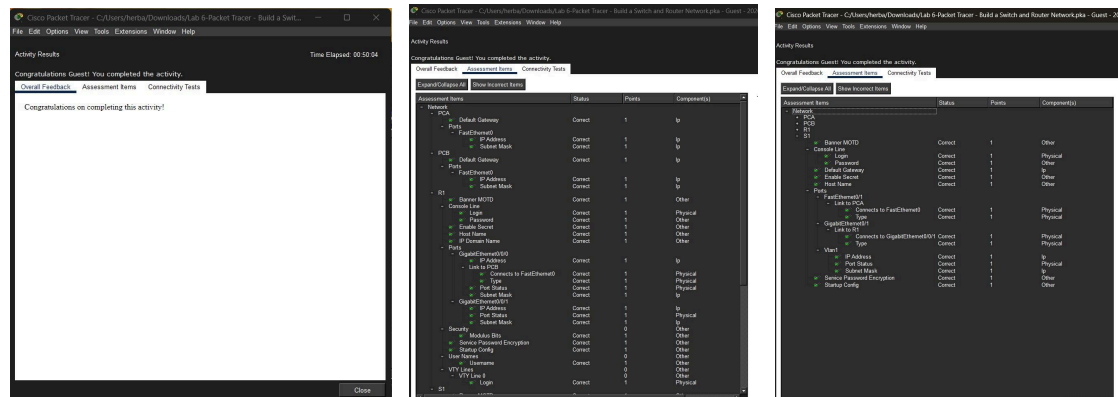


Packet Tracer Lab 6 - Build a Switch and Router Network



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0/0	192.168.0.1	255.255.255.0	N/A
	G0/0/1	192.168.1.1	255.255.255.0	N/A
S1	VLAN 1	192.168.1.2	255.255.255.0	192.168.1.1
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1
PC-B	NIC	192.168.0.3	255.255.255.0	192.168.0.1

Instructions

Part 1: Configure Devices and Verify Connectivity

In Part 1, you will set up the network topology and configure basic settings, such as the interface IP addresses, device access, and passwords. Refer to the Addressing Table at the beginning of this activity for device names and address information.

Step 1: Connect the devices.

The devices are already deployed in the workspace. You will connect them using the correct cables between the devices as listed below:

- Connect **PCA F0** to **S1 F0/1**.
- Connect **S1 G0/1** to **R1 G0/0/1**.
- Connect **R1 G0/0/0** to **PCB F0**.

Step 2: Assign static IP information to the PC interfaces.

- Configure the IP address, subnet mask, and default gateway settings on PC-A.
- Configure the IP address, subnet mask, and default gateway settings on PC-B.

- c. Ping PC-B from a command prompt window on PC-A.

Why were the pings not successful? **No**

Step 3: Configure R1.

- a. Console into the router and enable privileged EXEC mode. (**Hint:** Use console cable and terminal on a PC)
- b. Enter configuration mode. `config`
- c. Assign a device name to the router according to the Addressing Table.
- d. Assign **class** as the privileged EXEC encrypted password. Enable password class
- e. Assign **cisco** as the console password and enable login.
- f. Encrypt the plaintext passwords.
- g. Create a banner that warns anyone accessing the device that unauthorized access is prohibited.
- h. Configure the IP addresses according to the Addressing Table and activate both Ethernet interfaces on the router.
- i. Save the running configuration to the startup configuration file.

Were the pings successful? **Yes** Explain. **Now that the Router gigabit interfaces are configured correctly, it's able to connect to other devices.**

Step 4: Configure S1.

Note: Most of the commands on the switch are similar to the commands on the router in this step. Use the help (?) context as necessary.

- a. Console into the switch and enable privileged EXEC mode.
- b. Enter configuration mode.
- c. Assign a device name to the switch according to the Addressing Table.
- d. Assign **class** as the privileged EXEC encrypted password.
- e. Assign **cisco** as the console password and enable login.
- f. Encrypt the plaintext passwords.
- g. Create a banner that warns anyone accessing the device that unauthorized access is prohibited.
- h. Configure the IP address for the SVI for VLAN 1 according to the Addressing Table and activate the interface.
- i. Configure the default gateway according to the Addressing Table.
- j. Save the running configuration to the startup configuration file.

Part 2: Display Device Information

Step 1: Retrieve hardware and software information from the network devices.

- a. Use the **show version** command to answer the following questions about the router.

```
R1# show version
```

What is the name of the IOS image that the router is running? **Cisco IOS XE Software, Version 03.16.05.S - Extended Support Release**

- b. Use the **show version** command to answer the following questions about the switch.

```
S1# show version
```

What is IOS software image and version running on the switch? **C2960-LANBASE-M, SW version 12.2**

What is the model number of the switch? **WS-C2960-24TT**

Step 2: Display the routing table on the router.

Use the **show ip route** command on the router to answer the following questions.

```
R1# show ip route
```

What code is used in the routing table to indicate a directly connected network? **C**

How many route entries are coded with a C code in the routing table? **2**

What interface types are associated to the C coded routes? **gigabitEthernet 0/0/0 and 0/0/1**

Step 3: Display interface information on the router.

Use the **show interface g0/1** to answer the following questions.

```
R1# show interfaces g0/0/1
```

What is the operational status of the G0/0/1 interface? **GigabitEthernet0/0/1 is up, line protocol is up (connected)**

What is the Media Access Control (MAC) address of the G0/01 interface? **Hardware is Lance, address is 0060.3edc.7802**

How is the Internet address displayed in this command? **Internet address is 192.168.1.1/24**

Step 4: Display a summary list of the interfaces on the router and switch.

There are several commands that can be used to verify an interface configuration. One of the most useful of these is the **show ip interface brief** command. The command output displays a summary list of the interfaces on the device and provides immediate feedback to the status of each interface.

- a. Enter the **show ip interface brief** command on the router.

```
R1# show ip interface brief
```

```
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0/0 192.168.0.1 YES NVRAM up up
GigabitEthernet0/0/1 192.168.1.1 YES NVRAM up up
Serial0/1/0 unassigned YES unset down down
```

```
Serial0/1/1 unassigned YES unset down down
Vlan1 unassigned YES NVRAM administratively down down
```

- b. Enter the **show ip interface brief** command on the switch.

```
S1# show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/1 unassigned YES unset down down
FastEthernet0/2 unassigned YES unset down down
<output omitted>
GigabitEthernet0/1 unassigned YES unset up up
GigabitEthernet0/2 unassigned YES unset down down
Vlan1 192.168.1.2 YES manual up up
```

Part 3: Secure Remote Access to the Router

Step 1: Set the IP domain name and generate secure keys.

- a. On R1, configure the domain name as **academy.net**.

```
R1(config)# ip domain-name academy.net
```

- b. Generate RSA keys with a **1024** key length.

```
R1(config)# crypto key generate rsa
```

The name for the keys will be: R1.academy.net

Choose the size of the key modulus in the range of 360 to 2048 for your General Purpose Keys. Choosing a key modulus greater than 512 may take a few minutes.

```
How many bits in the modulus [512]: 1024
```

```
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]
```

Step 2: Create an SSH user and configure VTY lines for SSH-only access.

- a. Create a user with **SSHuser** as the username and **cisco** as the secret password.

```
R1(config)# username SSHuser secret cisco
```

- b. Configure the VTY lines to use the local username database for login credentials.

```
R1(config)# line vty 0 4
```

```
R1(config-line)# login local
```

- c. The VTY lines should only allow SSH for remote access.

```
R1(config-line)# transport input ssh
```

Step 3: Verify SSH Implementation.

- a. Click PCA, select **Command Prompt** in the Desktop tab.
- b. At the prompt, enter **ssh -l SSHuser 192.168.1.1**.

c. Enter **cisco** when prompted for the password.

What is the displayed message? **Unauthorized access prohibited. Username and Password required!**

Reflection

1. If the G0/0/1 interface showed administratively down, what interface configuration command would you use to turn the interface up? **No shutdown**

2. What would happen if you had incorrectly configured interface G0/0/1 on the router with an IP address of 192.168.1.2? **Configuring R1 G0/0/1 as 192.168.1.2 would**

- **create a duplicate IP conflict with S1 and**
- **prevent proper routing between networks.**
- **PC-A would lose access to its correct default gateway, causing connectivity and SSH issues.**